

Exercise 5.2

The formula for converting a temperature from Fahrenheit to Celsius is:

$$c = (5/9)*(f - 32)$$

and Celsius to Fahrenheit

$$f = (9/5)*c + 32$$

where f is the temperature in Fahrenheit and c is the temperature in Celsius.

Copy and paste the following into **main.py**. NOTE: You will have to update the function signature for `convert_temp`.

```
def convert_temp():
    pass

if __name__ == "__main__":
    # I would comment some of these out as you build up parts of convert_temp
    # e.g. just start by getting convert_temp(20) to work
    print(convert_temp(20)) # Prints 68.0
    print(convert_temp(-10)) # Prints 14.0
    print(convert_temp(68, Fahrenheit=True)) # Prints 20.0
    print(convert_temp(-100, Fahrenheit=True)) # Prints -73.33
```

Write a function that:

- Converts Celsius to Fahrenheit. The Celsius temperature is passed as a parameter and it returns the corresponding Fahrenheit temperature.
- Convert Fahrenheit to Celsius. The Fahrenheit temperature is passed as a parameter and it returns the corresponding Celsius temperature.
- Return **None** if the input for Celsius is below absolute zero, -273.15°C.
- Return **None** if the input for Fahrenheit is below absolute zero, -459.67°F.

This should be done with a keyword argument **Fahrenheit**

```
convert_temp(20) # convert celsius to fahrenheit. Returns 68.0
convert_temp(-10) # convert celsius to fahrenheit. Returns 14.0
convert_temp(68, Fahrenheit=True) # convert fahrenheit to celsius. Returns 20.0
convert_temp(-100, Fahrenheit=True) # convert fahrenheit to celsius. Returns
-73.33
```

You should round your answer to 2 decimal places using the **round** function. e.g. `round(3.14159, 2)` is **3.14**.

To pass all the tests, your function should also take into account absolute zero, which is -273.15°C , or -459.67°F and return `None` in either of these cases.

```
convert_temp(-280) # convert celsius to fahrenheit. Returns None
convert_temp(-460, Fahrenheit=True) # convert fahrenheit to celsius. Returns None
```

I would leave this until you have the main conversion working then add this functionality.

I would also suggest that you test your function by printing out the result of calling it with some of the values above. e.g.

```
print(convert_temp(20)) # should print 68.0 out to the console
```

Remember that the function needs to be above any print statement otherwise you will get a `NameError`.
